

Mark Wilson, *Physics Avoidance*, Essay 8: Semantic Mimicry

0. Introduction: The “Saloon Façade” and Physics Avoidance

Scientific theories are often presented as complete, standalone descriptions of “possible worlds,” but in practice they function more like movie-set facades (e.g. an Old West saloon). They are propped up by “hidden machinery” to be complete only for specific modelling purposes; attempting to model every degree of freedom or microscopic collision would render the governing equations mathematically useless (p. 337).

- **The “Sheriff of Bullet Valley” Theme:** In this analogy, the “Laws of Nature” (interior equations) take all credit for explanatory success, while an “unnoticed gaggle of helpful nephews”—boundary conditions, Green’s functions, interfacial jumps—secretly performs the critical labour (p. 331). The key philosophical problem: explanatory credit is systematically misattributed on the Theory T picture.
- **Physics Avoidance:** The practice of “skirting difficult calculations” or “deductive evasion” to reach predictive success via clever mathematical stratagems (p. 326). Wilson’s claim is not that this is illegitimate—it is often the only viable route—but that it is philosophically invisible on the standard account.
- **Semantic Mimicry:** Situations where researchers believe their linguistic expressions signify one physical underpinning when the modelling’s success actually relies on entirely different supportive structures (pp. 326–327). The key question: does successful prediction validate the justificatory story told about it?

The positive account Wilson gestures toward requires a richer vocabulary for the “hidden machinery”: extension elements, Green’s functions, weak solutions, and shared-space constructions each deserve conceptual recognition in their own right, not merely as approximation aids.

1. Critique of “Theory T” Thinking and the Initial Conditions “Grab Bag”

Philosophers often use a “broad brush” that erases critical structural distinctions by treating all auxiliary data as an undifferentiated pile labelled “initial conditions” (pp. 337–338).

- **Temporal Distortion:** Mislabeled boundary conditions actually extend into the future and often involve their own complex differential equations (e.g. gas diffusion through a bubble wall) (p. 338). The grab-bag label conceals that these “conditions” may themselves encode substantive physics.
- **Proper Significance:** Technically, “initial conditions” refers strictly to data specified at a single moment t_0 , a paradigm derived from Newtonian celestial mechanics where no external boundaries are considered (p. 338).
- **The “Illusion of ODE Expectation”:** The false belief that all of physics follows the simple evolutionary template of planetary motion (Ordinary Differential Equations), ignoring the distinct complexities of continuum mechanics (Partial Differential Equations) (p. 340).

The ODE template for an initial-value problem is structurally simple: given a state $x(t_0)$ and a rule $dx/dt = f(x, t)$, one integrates forward. PDEs break this template fundamentally. For a boundary-value problem over a region Ω with data on $\partial\Omega$, there is no analogous “marching” procedure, and the “conditions” are not localised in time at all. The philosophical problem: the initial conditions grab-bag obscures a structural distinction that makes a difference to what kind of explanation is possible.

2. Extension Elements: Green’s Functions and Analytic Informativeness

“Original modelling borders” are defined by classical standards of smoothness and differentiability. “Extension elements” allow physical reasoning to continue when the sanctioned model encounters a roadblock, such as a sharp corner (pp. 328–329).

- **Green’s Functions as Extension Elements:** These are “out-of-territory” because they contain singularities—they represent “singular tugs” or point-loads. A Green’s function $G(x, x')$ is defined as the solution to the equation with a point source (Dirac delta) at x' (pp. 329–330):

$$L[G(x, x')] = \delta(x - x') \text{ (defining equation for } G\text{)}$$

Given a Green’s function for a linear operator L , the solution to the full problem $L[u] = f$ is obtained by superposition:

$$u(x) = \int G(x, x') f(x') dx' \text{ (integral representation)}$$

- **The “Kernel” as a Smearing Function:** The Green’s function acts as a kernel inside the integral. Multiplying the singularity by the smoother load $f(x')$ “smears” it enough to satisfy the physical equations. The singularity is not eliminated but controlled (p. 330).
- **Analytic vs. Numerical Data:** Analytic formulas are preferred because they preserve structural features and symmetries of the equations—i.e. “algebraic information” about how solutions depend on parameters. Numerical simulations produce approximate trajectories that can diverge and lose these underlying functional relationships (pp. 330–331). This preference is not merely aesthetic: it tracks what explanatory work a solution can do.

3. Technical Case Study 1: Hilbert’s Direct Method (Completion)

Mathematicians can “fill in the holes” of a model using functional analysis to reach non-classical solutions (pp. 346–347).

- **The Bang-Bang Solution:** In optimal planning (e.g. rowing to an island), the “best” path may be a “broken line” with a sharp corner that technically lacks a second derivative (p. 346). Classical variational calculus cannot handle this: the relevant functional is not differentiable at the solution.
- **Completion and the Sobolev Space:** Hilbert’s method passes to a larger function space—a Sobolev-type space—in which limits of smooth functions with bounded energy are admitted as valid solutions, even when non-smooth. Formally, we work in a space H where sequences u_n with

$$\int |u_n|^2 + |\nabla u_n|^2 \leq C$$

have convergent subsequences. The sharp-cornered path is the limit of smooth paths in this topology (p. 347).

- **The Topological Mesh / “Haughty General”:** The smooth paths act as “aides-de-camp” that advise the “haughty general” (the sharp-cornered path) on its energy value. The metaphor makes vivid that the completion is not a mere trick: the non-classical solution inherits its properties from the smooth ones that surround it (p. 347).

The philosophical point: the “completion” is a substantive extension of the modelling landscape, not just a notational convenience. What counts as a “solution” has shifted—and with it, what counts as an explanation.

4. Technical Case Study 2: The Rayleigh–Ritz Method (Subspace Search)

This provides the underlying rationale for the Finite Element Method (FEM) used by aircraft engineers (pp. 354–355).

- **Search within a Subspace:** Instead of viewing finite elements as physical “little plates” welded together, we view them as trial functions within a restricted subspace S^{FE} of a larger function space. We seek:

$$u^* = S^{FE} \text{ such that } E[u^*] = \min_u \{ S^{FE} \} E[u] \text{ (subspace minimisation)}$$

- **Energy Minimisation:** We search for the function within the restricted space that possesses the least energy. The restricted minimiser is the best approximation to the true solution available within S^{FE} (p. 355).
- **The Mimicry:** 1940s engineers reached successful results because their “dodgy” physical model of welding plates mirrored the subspace-search strategy formalised by Rayleigh–Ritz (p. 356). Crucially: the engineers’ predictive success gave them no internal signal that their justificatory story was wrong. This is Wilson’s sharpest case that successful prediction does not validate the account of why the model works.

Connection to the seminar’s EFT focus: integrating out heavy modes in an EFT similarly restricts attention to a low-energy subspace. The question is whether the restriction is justified by an underlying story (Wilsonian RG) or is itself an instance of mimicry—working for reasons that differ from what the formalism suggests.

5. Technical Case Study 3: Test Functions and the “Weak Solution”

Mathematical “distributions” can be linked directly to the physical act of measurement (pp. 349–350).

- **Test Functions $\theta(x)$:** These are mathematical models of measuring instruments. Because point-values are unmeasurable, we define a property by how it affects all measuring instruments (test functions) (p. 349). They are smooth, compactly supported functions—precisely those that can serve as idealised probes.
- **From Strong to Weak Form:** A “strong” solution satisfies $Lu = f$ pointwise. Multiplying by a test function θ and integrating by parts transfers derivatives from u to θ :

$$\int u \cdot L^*[\theta] dx = \int f \cdot \theta dx \text{ for all test functions } \theta \text{ (weak formulation)}$$

A “weak solution” satisfies only this integral identity. Because θ is smooth, the derivatives that u might lack are absorbed by $L^*[\theta]$. Sharp corners and discontinuities become admissible (p. 352).

- **Operational Definition:** Two regions possess the same property if they affect all measuring instruments in the same way (p. 350). This operationalises what it means for a distribution to “have” a value—an implicit philosophy of measurement embedded in the mathematical framework.

Note the connection to Section III: a Green’s function is itself a distribution, satisfying its defining equation only in the weak sense. The weak formulation provides the rigorous underpinning for why Green’s functions are “out-of-territory” extensions rather than classical solutions.

6. The Shared Space Metaphor and Semantic Mimicry

Shared space is a mathematical arena where conceptually distinct physical roles “cohabit” to facilitate calculation (p. 353).

- **Geometric Precision:** Modern mathematical physics distinguishes between different geometrical structures—vectors vs. one-forms—defined over a manifold. While they can be pushed into a shared vector space via a metric, they remain conceptually distinct (pp. 353–354). The shared space makes them look identical; the geometry makes them different.
- **Trial Solutions vs. Manipulations:** In FEM, “trial solutions” (guesses for the solution) and “manipulations” (variational tweaks) share a Banach space because they obey the same endpoint conditions (p. 354). Formally they are the same kind of object; physically they play entirely different roles.
- **The Hazard:** Because these roles look identical in the shared space, 1940s aircraft engineers mistook a mathematical manipulation for a direct physical interaction between metal plates. While the math worked, the “justificatory trappings”—the story of why it worked—were entirely wrong (p. 356). This is semantic mimicry in its purest form: the expression “the plates interact” succeeds predictively while signifying the wrong physical underpinning.

The central tension for discussion: if the math works whether or not the justificatory story is correct, what is at stake in getting the story right? Three possible answers deserve scrutiny:

- **Epistemic:** the wrong story will eventually mislead us when we try to extend or correct the model.
- **Metaphysical:** there is a fact of the matter about what the model “really” describes, independent of predictive success.
- **Semantic:** the concepts themselves (force, interaction, rigidity) have migrated in meaning, so the question of whether the story is “correct” is not well-formed without specifying which regime’s vocabulary we are using.

It is somewhat unclear where Wilson’s own diagnosis lies (cf. Discussion Questions below). Somewhat similarly, this connects directly to the seminar’s EFT questions: does the “integrate out” metaphor in EFT describe a genuine physical operation, or is it a shared-space artefact that mimics one?

7. Discussion Questions

1. Realist? To what extent?
 - a. Wilson says he is “[...] as confirmed a “scientific realist” as walks this planet and believe that ongoing science gradually accumulates a large set of reliable “truths”.” (fn. 57). But it is difficult to see (when we acknowledge *semantic mimicry*) what we should be realist about? Are we meant to be realist about the interior equations *and* the boundary conditions, Green’s functions (etc.) too?
2. How to change a theory?
 - a. In a familiar case (*Mercury’s perihelion*; NG→GR), we see a case of theory change where the underlying physics is being replaced. Wilson seems to identify a different case: we have cases where simply (and only) changing the applied math techniques—even when the underlying theory stays the same—can still lead to us a different set of conclusions about what is going on. What remains as ‘robust’? Do we place the “smearing” and “averaging” strategies as the focus?
3. How to derive phenomena?
 - a. As we had seen earlier in the seminar, B&W argue that scientists reliably derive phenomena from data through stable, repeatable inferential procedures — the phenomena so derived then serving as the proper explanandum for theory. But Wilson's semantic mimicry cases put pressure precisely on the reliability of these inferential procedures. If the inferential pathway from data to phenomenon can be a mimic — producing stable, reproducible results for reasons quite different from those the scientists take themselves to be exploiting — does this undermine B&W's confidence that phenomena are robustly “real” in virtue of being stably derivable?
 - i. Consider the explanations and descriptions that the aircraft engineers.

References

Wilson, M. (2017). *Physics Avoidance: And Other Essays in Conceptual Strategy*. Oxford University Press.